



AMTEC
AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0387



Agricultural Machinery – Heated-Air Mechanical Grain Dryer AMCC LSU-12T

Test Report Number	2023 – 0387
Date of Issuance	December 19, 2023
Valid Until	December 19 2028
Agricultural Machinery	Heated-Air Mechanical Grain Dryer
Type	Batch (Recirculating), Indirect-fired, Renewable, Tested for Rice
Brand	AMCC
Model	LSU-12T
Test Requested by	AGRICOMPONENT CORPORATION Tagaran, Cauayan City, Isabela

SUMMARY

Performance Criteria	Requirement as per PNS/PAES 201:2015	AMTEC Test
Moisture Gradient, %, maximum	2.0 ^a	1.80
Drying Rate, %/h	0.8 ^b	0.612
Product Quality ^c		
a) Cracked Grain, % increase, maximum	3.0 ^d	4.67
b) Head rice, % decrease, maximum	5.0	8.14
c) Hulled/ Damaged Grain, % increase, maximum	2.0	0
Drying Loss/Spillage, %, maximum	1.0	0
Drying Efficiency, %, minimum	75	40.86
Heating System Efficiency, %, minimum	50 ^e	93.22
Blower flow rate, cmm/kg (cmf/ton)	0.05 (1750)	0.030 (1045)
Static Pressure, mm H ₂ O (in H ₂ O)	25 (1)	43.34 (1.71)
Grain Holding Capacity, kg	12000 ^f	12011.5
Discoloration, Traces of Unburned Fuel, Ashes on Grain Surface	Shall not be present	None
Fermented or Musty Smell	Shall not be present	None
Thermometer	Shall be provided	Provided
Pressure Gauge	Shall be provided	Provided
Noise Level, dB(A), maximum	92	79.7

^a Requirement for recirculating heated-air mechanical grain dryer.

^b Requirement for batch type heated-air mechanical grain dryer for rice.

^c Allowable difference between the laboratory analysis and machine's performance.

^d Requirement for batch type heated-air mechanical grain dryer for rice.

^e Requirement for indirect-fired biomass fuel.

^f As specified by the manufacturer.

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Provided
e. Maintenance instruction	Provided
f. Storage	Provided
g. Handling, reception, transportation, assembly, and installation	Provided except for assembly
h. Specifications	Provided
i. Dismantling and disposal	Provided
j. Warranty	Provided
k. Parts list	Not provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	AGRICOMP MACHINERIES AND CONSTRUCTION CORP. Tagaran, Cauayan City, Isabela
4. Heat source	AMCC SBF Biomass Furnace
5. Safety controls/devices	a. Emergency stop button, individual switches for each component, and alarm and light indicator for power trip, are provided. b. The transmission system and other moving parts are fully covered. c. Access ladder with cage is provided.
6. Dust collection systems	A dust collection system is installed at the top portion of the holding bin.
7. Instrumentation, setting and adjustments	a. Main switch is provided. b. Temperature controller and adjustment knobs for moisture setting according to crop and moisture correction/setting were provided. c. A pressure gauge is provided on the plenum.
8. Loading and unloading	a. The wet paddy is manually loaded on the feeding table with hopper which is connected on a screw conveyor. The paddy rice is then brought to the drying bin through a bucket elevator. Feeding of input can be controlled through a sliding gate in the hopper. b. The dried paddy is taken to the bagging bin through the pipe connected to the bucket elevator.

OBSERVATIONS (Cont'n)

9. Number of operators needed to load the grains	Three (One loader and two haulers)
10. Number of operators needed to load the biomass (rice hull)	Two (One loader and one hauler)
11. Shall have adequate provision for fire control	The dryer is equipped with flame detector.
12. Shall have adequate protection from or for all moving parts	The moving parts are provided with covers.
13. Shall be provided with features for access to parts during repair, maintenance, and operation	All parts are accessible for repair, maintenance, and operation.
14. The noise emitted by the heated-air mechanical dryer shall not exceed 92 dB(A).	The maximum noise level measured was 79.7 dB(A) taken at the feeding operator's ear level.
15. Provision shall be made for dust control. The dust emission of the machine shall be within the acceptable limits set by the Department of Environment and Natural Resources.	The machine is provided with a dust aspirator located at the top of the drying bin and a cyclone outside the warehouse. However, dust emission was not verified due to limited technical capability on site.
16. Shall be free from defects that may be detrimental to its use	The machine is free from any manufacturing defects that may be detrimental.
17. Shall be free from sharp edges and surfaces that may hurt the operator	The machine is free from any sharp edges or surfaces that may injure the operator.
18. All metal parts should be machine bent, pressed and cut to avoid rough surfaces. All rough surfaces should be machine finished and smoothed.	Metal parts were machine bent. There were no rough surfaces observed on the machine.
19. The warning notice shall be in accordance with PNS/BAFS 330:2022 – Technical Means for Ensuring Safety – Guidelines.	Warning notices in text form are located on the control panel and grain moisture meter.
20. Uniformity of parts and components for same brand and model must be maintained.	The dryer and biomass furnace has the same brand (AMCC). However, their corresponding parts and components were of different brands and models.

OBSERVATIONS (Cont'n)

21. Provision shall be made so that it can withstand extreme weather conditions to which it can be exposed.	The dryer was installed inside a permanent structure.
22. Other remarks	a. An operator's manual applicable for 12 tons dryers but not specific to the model tested was submitted by the requesting party. b. The dryer system was connected to a generator for source of electricity.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	August 6-7, 2023
Location of Test	Brgy. Lactawan, Solano, Nueva Viscaya 16°33'31.73" N, 121°11'17.75" E
Items Tested	Operator's Manual, Drying Capacity, Moisture Reduction Rate, Heating System Efficiency, Drying Efficiency, Drying System Efficiency, Furnace Efficiency, Furnace Burning Efficiency, Moisture Content Variation, Moisture Content Gradient, Fuel Consumption Rate, Power Consumption, Number of Operators, and Quality of Dried Grains

METHODS OF TEST

The grain dryer was tested based on the PNS/PAES 202:2015 Agricultural Machinery – Heated-air Mechanical Grain Dryer – Methods of Test.

DESCRIPTION OF THE MACHINE



Figure 1. Different parts of the AMCC LSU-12T Heated-Air Mechanical Grain Dryer and its components heated by AMCC SBF Biomass Furnace include the following: (a) Biomass furnace, (b) Bagging bin, (c) Elevator, (d) Drying bin, (e) Access ladder with cage, (f) Control panel, (g) Feeding table with hopper, and (h) Moisture meter.

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification
1	Drying rate, %/h	0.8-2.0	0.612
2	Overall dimensions, mm (installed)		
2.1	Length	5320	5088
2.2	Width	6190	5920
2.3	Height	8880	8840
3	Drying bin		
3.1	Type	Grain-circulation	Vertical, rectangular
3.2	Dimensions, mm		
3.2.1	Length	ND	2340
3.2.2	Width	ND	1925
3.2.3	Height	ND	7800
3.3	Holding capacity, kg	Rice: 12000; Corn: 15000	12011.5
3.4	Maximum grain depth, mm	ND	NA
3.5	Material		
3.5.1	Frame	ND	204 Stainless steel ^b
3.5.2	Holding bin	ND	204 Stainless steel ^b
3.5.3	Wall	ND	204 Stainless steel ^b
4	Fan		
4.1	Type	ND	Axial
4.2	Brand and model	ND	ND ^c
4.3	Air flow rate, m ³ /min	ND	355.27
4.4	Static pressure, Pa	ND	425.05
4.5	Material	ND	ND ^d
4.6	Prime mover	ND	1.1 kW WEG Electric motor
5	Heater		
5.1	Type	Biomass furnace	Biomass furnace
5.2	Brand and model	AMCC SBF	AMCC SBF ^a
5.3	Fuel	Rice hull/corn cob	Rice hull
5.4	Heat output, kJ/h	ND	536027.56
5.5	Fuel consumption rate, kg/h	50-130	38.49

^a The data were obtained from the component's name plate and operator's manual provided by the requesting party.

^b Determined using BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer. The percent elemental composition is summarized in Table 1.

^c The component does not have a name plate during testing.

^d The component was not accessible during testing.

Note: NA – Not Applicable, ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
5.6	Capacity of fuel tank, L	ND	NA
5.7	Mechanism for fuel feed	ND	With feeder
5.8	Method of temperature control	ND	Digital
5.9	Material of construction	ND	Metal sheet and bricks
6	Material handling system		
6.1	Feeding table with hopper		
6.1.1	Dimension, L × W, mm	ND	1265 × 1020
6.1.2	Height from the ground, mm	ND	1015
6.1.3	Material of construction	ND	204 Stainless steel ^b
6.1.4	Other features	ND	Feeding can be controlled through the sliding gate
6.2	Elevator		
6.2.1	Number	ND	One
6.2.2	Height, m	ND	8840
6.2.3	Capacity, t/h	ND	NM ^c
6.2.4	Belt dimensions, W × t, mm	ND	170 × 5.33 ^c
6.2.5	Elevator buckets		
6.2.5.1	Dimension, L × W × H, mm	ND	147 × 117 × 100
6.2.5.2	Material of construction	ND	Plastic
6.2.6	Drive motor		
6.2.6.1	Type	Electric motor	Electric motor
6.2.6.2	Rated power, kW	2.2	2.2 ^a
6.2.6.3	Rated speed, rpm	1725	1725 ^a
6.2.6.4	Phase	Three-phase	Three-phase
6.2.6.5	Current, A	8.79/5.09/4.40	8.79/5.09/4.40 ^a
6.2.6.6	Voltage, V	220/380/440	220/380/440 ^a
6.2.6.7	Frequency, Hz	60	60 ^a

^a The data were obtained from the component's nameplate. The requesting party did not submit complete machine specifications.

^b Determined using BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer. The percent elemental composition is summarized in Table 1.

^c Not verified due to installation.

Note: NA – Not Applicable, ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
6.3	Auger		
6.3.1	Drive motor		
6.3.1.1	Type	Electric motor	Electric motor
6.3.1.2	Rated power, kW	1.1	1.1 ^a
6.3.1.3	Rated speed, rpm	1750	1750 ^a
6.3.1.4	Phase	Three-phase	Three-phase
6.3.1.5	Current, A	4.60/2.66/2.30	4.60/2.66/2.30 ^a
6.3.1.6	Voltage, V	220/380/440	220/380/440 ^a
6.3.1.7	Frequency, Hz	60	60 ^a
6.4	Blower		
6.4.1	Drive motor		
6.4.1.1	Type	Electric motor	Electric motor
6.4.1.2	Rated power, kW	1.1	1.1 ^a
6.4.1.3	Rated speed, rpm	1750	1750 ^a
6.4.1.4	Phase	Three-phase	Three-phase
6.4.1.5	Current, A	4.60/2.66/2.30	4.60/2.66/2.30 ^a
6.4.1.6	Voltage, V	220/380/440	220/380/440 ^a
6.4.1.7	Frequency, Hz	60	60 ^a
6.5	Biomass furnace		
6.5.1	Drive motor		
6.5.1.1	Type	Electric motor	Electric motor
6.5.1.2	Rated power, kW	0.75	0.75 ^a
6.5.1.3	Rated speed, rpm	1730	1730 ^a
6.5.1.4	Phase	Three-phase	Three-phase
6.5.1.5	Current, A	3.12/1.81/1.56	3.12/1.81/1.56 ^a
6.5.1.6	Voltage, V	220/380/440	220/380/440 ^a
6.5.1.7	Frequency, Hz	60	60 ^a
6.6	Output chute		
6.6.1	Dimension, L × W, mm	ND	115 × 125
6.6.2	Height from the ground, mm	ND	810
6.6.3	Material of construction	ND	204 Stainless steel ^b

^a The data were obtained component's nameplate. The requesting party did not submit complete machine specifications.

^b Determined using BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer. The percent elemental composition is summarized in Table 1.

Note: ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification^a	AMTEC Verification
6.6.4	Other features	Provided with diesel-fed burner	a. Discharge rate and loading rate can be controlled through the sliding gate. b. The dryer is also equipped with diesel-fed burner.
7	Instruments and controls		
7.1	Moisture content	ND	Digital
7.1.1	Location	ND	Below the hopper drying bin
7.2	Drying air temperature	ND	Digital
7.2.1	Location	ND	At the plenum
7.3	Static pressure	ND	None
7.3.1	Location	ND	NA
7.4	Grain temperature	ND	Digital
7.4.1	Location	ND	At the plenum

^a The requesting party did not submit complete machine specifications.

Note: NA – Not Applicable, ND – No Data

SPECIFICATIONS (Cont'n)

Item	Manufacturer's Specification ^a	AMTEC Verification
8 Safety features		
8.1 Product safety	a. Fire hazard warning b. Flame sensor	a. Emergency stop button b. Drying air and grain temperature sensors c. Moisture meter d. Fire hazard warning e. Flame sensor
8.2 Machine safety	ND	Circuit breaker, over/under voltage relays
9 Operator safety	a. Fire hazard warning b. Flame sensor	a. Covers for the belt guard, transmission system, and biomass furnace b. Fire hazard warning c. Flame sensor d. Dust collection system e. Warning signs f. Railings at the top of the dryer.

^a The requesting party did not submit complete machine specifications.

Note: ND – No Data

RESULTS

Item		Data
1	Performance test	
1.1	Initial weight of test material, kg	12011.5
1.2	Final weight of test material, kg	10182.0
1.3	Initial moisture content, % _{wb}	26.79
1.4	Final moisture content, % _{wb}	13.63
1.5	Actual drying time, h	21.5
1.6	Drying capacity, kg/h	558.67
1.7	Rate of moisture removal, kg/h	85.09
1.8	Moisture reduction rate, %/h	0.612
1.9	Average drying air temperature, °C	59.4
1.10	Ambient air condition	
1.10.1	Average dry bulb temperature, °C	37.6
1.10.2	Average wet bulb temperature, °C	30.6
1.10.3	Relative humidity, %	60.6
1.11	Exhaust air condition	
1.11.1	Average dry bulb temperature, °C	38.3
1.11.2	Average wet bulb temperature, °C	37.7
1.11.3	Relative humidity, %	68.3
1.12	Average grain temperature, °C	29.4
1.13	Fan air velocity, m/s	13.37
1.14	Total air flow rate, m ³ /min	355.27
1.15	Plenum static pressure, mmH ₂ O	43.34
1.16	Biomass furnace fuel consumption, kg/h	38.49
1.17	Power consumption	
1.17.1	Total electrical input power, kW	14.71
1.17.2	Line voltage, ACV	221.0
1.17.3	Load current, A	38.4
1.18	Noise level at operator's ear level, dB(A)	
1.18.1	Feeding operator	79.7
1.18.2	Machine operator	79.2
1.18.3	Bagger	78.3
1.19	Heat utilization, kJ/kg water removed	5872.40
1.20	Heat system efficiency, %	93.22
1.21	Drying efficiency, %	40.86
1.22	Drying system efficiency, %	38.09

RESULTS (Cont'n)

Item		Data	
1.23	Moisture content variation at the end of drying, %	0.593	
1.24	Moisture content gradient, %	1.80	
1.25	Furnace efficiency, %	3.88	
1.26	Furnace burning efficiency, %	88.41	
2	Condition of test material	Control	Dryer Test
2.1	Variety	NSIC Rc 222	
2.2	Source	Nueva Vizcaya	
2.3	Paddy analysis		
2.3.1	Grain size, brown rice, mm		
2.3.1.1	Length	7.28	7.67
2.3.1.2	Width	2.02	2.03
2.3.1.3	Thickness	1.60	1.20
2.3.2	Grain type	Long and Slender	Long and Slender
2.3.3	Paddy bulk density, kg/m ³	528.84	550.34
2.3.4	Purity, %	99.57	99.69
2.3.5	Cracked, %	1.67	6.33
2.3.6	Brown rice		
2.3.6.1	Immature, %	14.23	14.67
2.3.6.2	Chalky, %	4.94	3.16
2.3.6.3	Yellow and fermented, %	2.34	2.27
2.3.6.4	Red rice, %	0	0.040
3	Laboratory milling test ^a		
3.1	Total milling recovery, %	62.69	66.63
3.2	Head rice recovery, %		
3.2.1	Based on milled rice	91.87	83.73
3.2.2	Based on rough rice	57.59	55.79
3.3	Broken rice, %		
3.3.1	Based on milled rice	6.48	12.72
3.3.2	Based on rough rice	4.06	8.48
3.4	Brewer's rice, %		
3.4.1	Based on milled rice	1.653	3.55
3.4.2	Based on rough rice	2.21	2.36

^a Satake laboratory test.

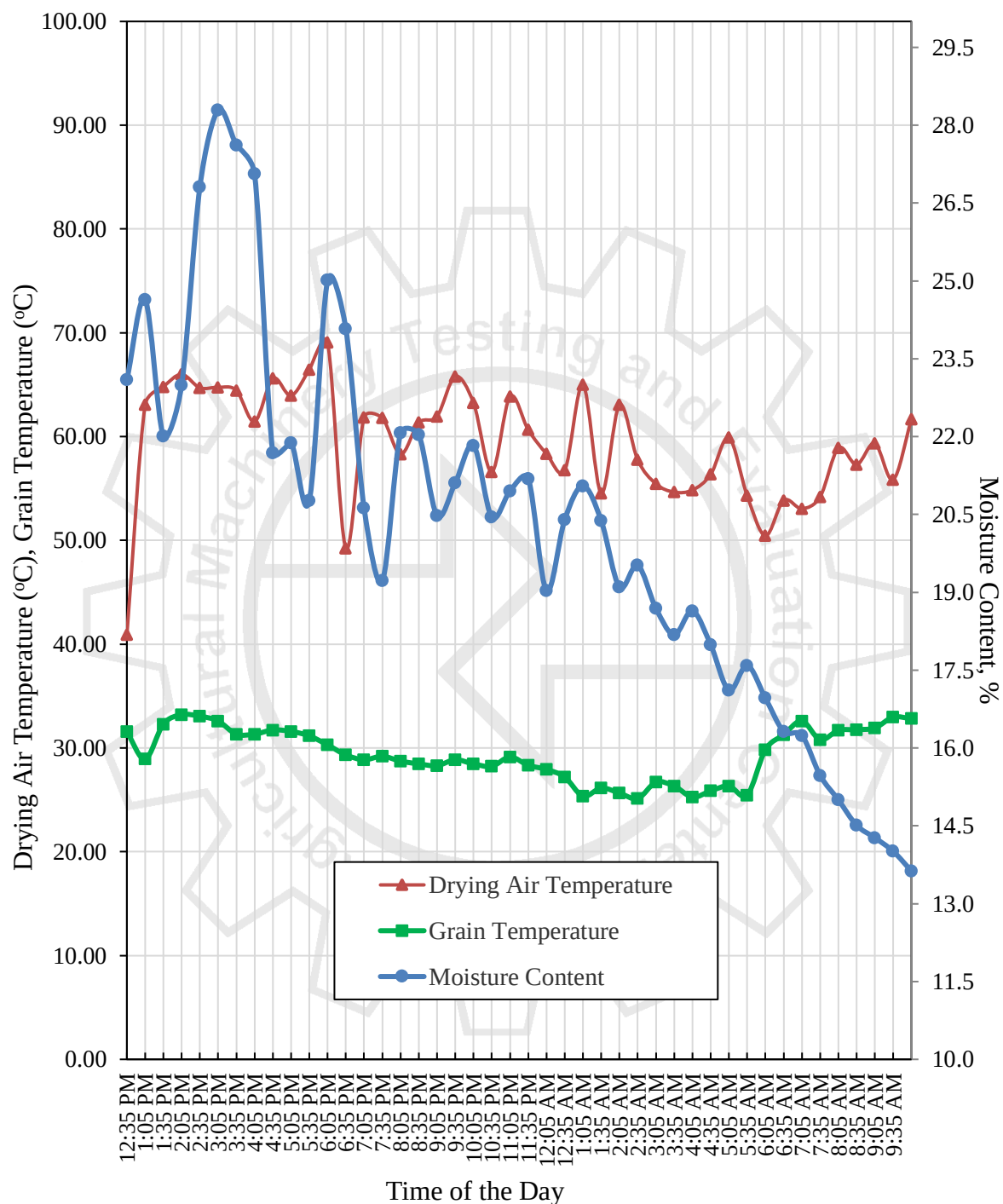
RESULTS (Cont'n)

Figure 2. Drying curve of the AMCC LSU-12T Heated-Air Mechanical Grain Dryer.

RESULTS (Cont'n)

Table 1. Elemental composition of the different components of AMCC LSU-12T Heated-Air Mechanical Grain Dryer

Element	Percent Composition^a		
	Frame, holding bin, and wall	Input loading hopper	Output chute
Titanium	< LOD	< LOD	< LOD
Vanadium	0.11	0.13	0.11
Chromium	-	-	-
Manganese	-	-	-
Iron	74.83	74.82	74.78
Cobalt	-	-	-
Nickel	-	-	-
Copper	0.22	0.21	0.19
Grade	204 Stainless Steel	204 Stainless Steel	204 Stainless Steel

^a Determined using the Bruker S1 Titan 800 X-Ray Fluorescence Analyzer.

Note: LOD – Limit of Detection

OBSERVATIONS



Figure 3. Markings and labels on the control panel of the AMCC LSU-12T Heated-Air Mechanical Grain Dryer.

OBSERVATIONS (Cont'n)

Figure 4. Nameplate of the dryer (top) and biomass furnace (bottom)

OBSERVATIONS (Cont'n)

Figure 5. Nameplate of the biomass furnace motor.

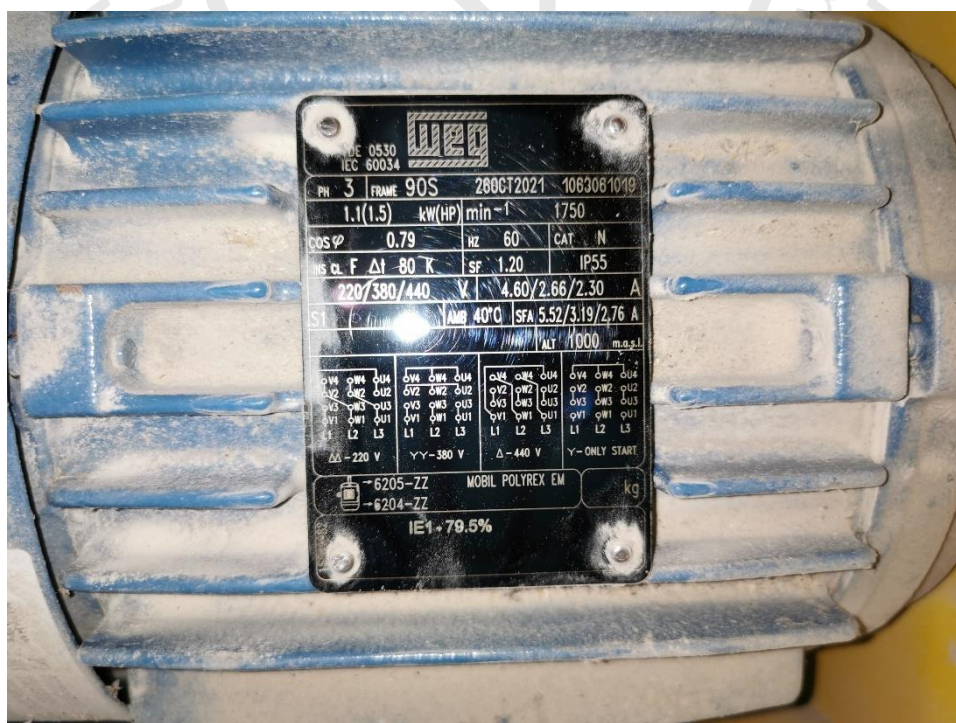


Figure 6. Nameplate of the blower motor.

OBSERVATIONS (Cont'n)

Figure 7. Nameplate of the elevator motor.

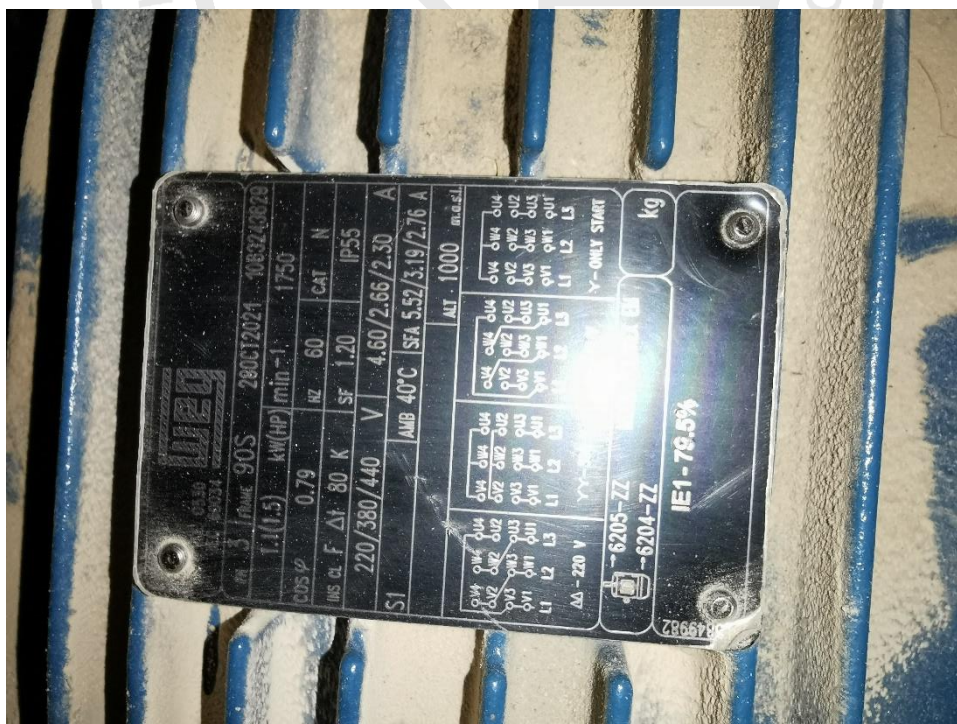


Figure 8. Nameplate of the auger motor.

OBSERVATIONS (Cont'n)

Figure 9. The generator used during the test.

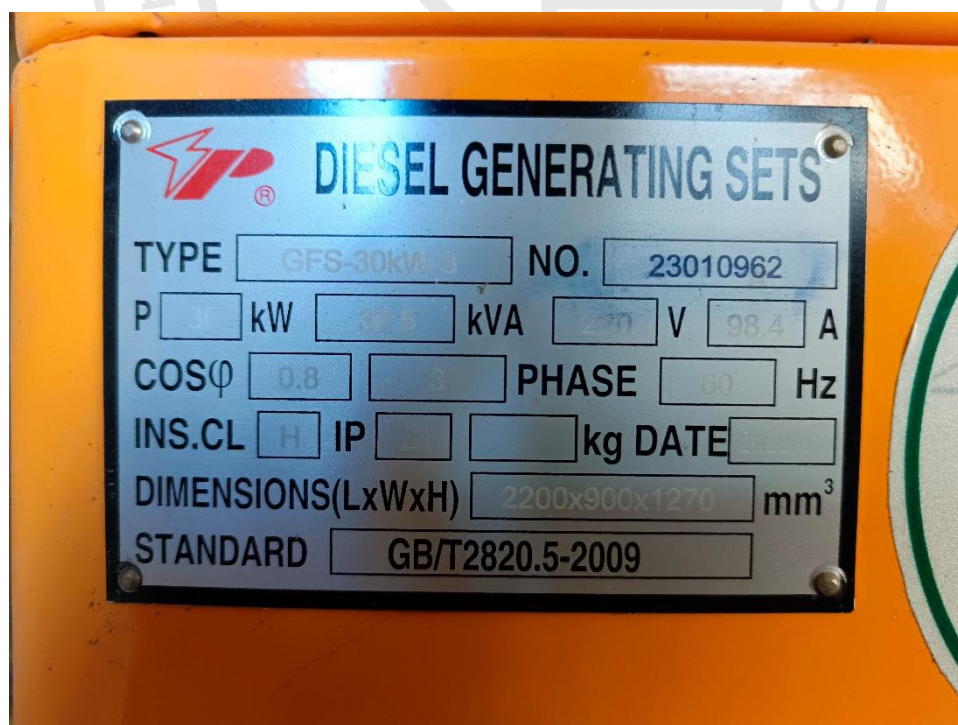


Figure 10. The generator used during the test.

Tested by:

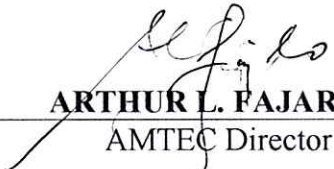

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